CYS 331 —Unit 2: Wireless Threats

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Learning Outcomes

- Classify major wireless threats and explain how attackers exploit open transmission environments.
- Distinguish passive threats from active threats and connect each to security objectives.
- Explain how interference, spoofing, traffic analysis, and weak cryptography undermine trust.
- Relate wireless security standards to practical defenses against common threat categories.

Kinds of Security Breaches

- Wireless breaches target confidentiality, integrity, availability, authentication, or accountability in different ways.
- Passive breaches observe communications silently, while active breaches modify or disrupt systems.
- Attackers often combine technical weaknesses with human mistakes and poor configuration.
- Understanding breach categories helps defenders choose suitable monitoring and preventive controls.

Eavesdropping and Traffic Analysis

- Eavesdropping captures signals in transit when protection is weak or absent.
- Traffic analysis studies patterns, timing, volume, and endpoints even when content stays hidden.
- Sensitive information may leak through metadata, behavior patterns, or repeated transmission habits.
- Encryption reduces exposure, but network design and user discipline still matter.

Communication Jamming and Radio Frequency

- Communication jamming intentionally overwhelms channels to deny reliable wireless service.
- Radio frequency interference may be accidental, environmental, or caused by nearby devices.
- Both problems reduce availability by degrading signal quality, throughput, and connection stability.
- Detection requires spectrum awareness, site surveys, redundancy, and adaptive channel planning.

Covert Channels and Denial of Service Attacks

- Covert channels secretly transfer information using unauthorized wireless pathways or timing tricks.
- Denial of service attacks exhaust resources, disconnect clients, or overload access points.
- Attackers exploit management frames, weak protections, and poor capacity planning.
- Resilience depends on monitoring, segmentation, rate limiting, and rapid incident response.

Spoofing and Theft of Services

- Spoofing imitates trusted devices, network names, or identities to deceive users.
- Theft of services occurs when unauthorized users consume connectivity without permission.
- Rogue access points and cloned identifiers often support impersonation attacks.
- Strong authentication and user awareness reduce opportunities for fraudulent access.

Cryptographic Threats and Weak Standards

- Weak algorithms, reused keys, and poor implementation undermine wireless confidentiality.
- Legacy protections fail when attackers capture enough traffic for practical analysis.
- Secure standards matter because strong design raises attack cost significantly.
- Defenders must retire obsolete protocols instead of trusting historical compatibility.

Wireless Security Standards and Defensive Mapping

- Security standards define authentication, encryption, integrity, and management expectations for deployments.
- Stronger standards protect against eavesdropping, impersonation, and key-recovery attacks better.
- Standards alone are insufficient without correct configuration, updates, and monitoring.
- Defenders should map each standard to the threat it is meant to resist.

Unit 2 Summary

- Wireless threats target secrecy, trust, service availability, and honest resource use.
- Passive observation and active disruption require different defensive strategies and tools.
- Spoofing, interference, traffic analysis, and weak cryptography often reinforce each other.
- Strong standards, correct configuration, monitoring, and awareness form the best defense.